

Date	04 July 2026
Team ID	To Be Assigned
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Soil & Weather Data Input	Collects soil nutrients (N, P, K), temperature, humidity, pH, and rainfall from the user.	Frontend UI	Done	0.5
2	Input Validation	Validates user inputs before sending them for prediction.	Frontend + Backend	Done	0.5
3	Data Preprocessing	Processes and prepares input data for the Machine Learning model.	ML Preprocessing	Done	1.0
4	Crop Recommendation	Predicts the most suitable crop using the trained Machine Learning model.	Prediction Engine	Done	1.0
5	Machine Learning Model	Trains and evaluates the crop prediction model for accurate recommendations.	Training Module	Done	0.5
6	Result Display	Displays the recommended crop in a clear and user-friendly format.	Frontend	Done	0.5
7	Model Loading	Loads the trained .pkl model during application startup..	Backend	Done	0.5
8	Error Handling	Handles invalid inputs and runtime exceptions gracefully.	Application Layer	Done	0.5